C# PRACTICAL ASSIGNMENT

Chapter 1 – C# & .NET Core | Hands-on Console Programming

Instructions

Create a separate C# console application for each question. Read the requirements carefully, accept the required input from the user, and display the expected output. Use the programming concepts covered in Chapter 1.

Part 1: Basic Programming and Operators

1. Student Result Calculator

Create a console application that accepts a student's name and marks in three subjects.
Requirements:
• Accept the student's name.
• Accept marks for three subjects.
• Calculate the total and average marks.
• Display the student's name, total and average.

```
Enter student name: Rahul
Enter marks in Subject 1: 85
Enter marks in Subject 2: 90
Enter marks in Subject 3: 75

Student Name: Rahul
Total Marks: 250
Average: 83.33
```

2. Electricity Bill Calculator

Create an application that calculates an electricity bill based on the number of units consumed.
Billing rates:

Units consumed	Rate per unit
First 100 units	■2
Next 100 units	■3
Above 200 units	■5

Requirements:
• Accept the number of units consumed.
• Calculate the bill based on the slabs.
• Display the total electricity bill.

3. Shopping Discount Calculator

Create a shopping bill application.
Requirements:
• Accept the product name and price.
• Accept the quantity.
• Calculate the total amount.
• If the total amount is above ■5,000, provide a 10% discount.
• Otherwise, no discount is applied.
• Display the final amount.

Part 2: Conditional Statements and switch

4. ATM Withdrawal Simulator

Create a simple ATM withdrawal application.
Requirements:
• Set an initial account balance of ■10,000.
• Accept the withdrawal amount.
• Check whether the withdrawal amount is positive.
• Check whether the account has sufficient balance.
• If sufficient, deduct the amount.
• Display the remaining balance.

```
Account Balance: 10000
Enter withdrawal amount: 3000

Withdrawal Successful!
Remaining Balance: 7000
```

5. Movie Ticket Booking

Create a simple movie ticket booking application.
Available categories:

Choice	Category	Price
1	Regular	■150
2	Premium	■250
3	VIP	■400

Requirements:
• Accept the user's choice using **switch**.
• Accept the number of tickets.
• Calculate the total price.
• Display the booking summary.
• Display an appropriate message for an invalid category.

6. Employee Salary Calculator

Create an application that calculates an employee's salary based on their designation. Use **switch**.

Designation	Basic salary
Developer	■30,000
Tester	■25,000
Manager	■45,000
Intern	■12,000

Requirements:
• Accept the employee name.
• Accept the designation.
• Use **switch** to identify the basic salary.
• Calculate an additional 10% bonus.
• Display the final salary.

Part 3: Loops – Real-world Applications

7. Multiplication Table Generator

Create a program that accepts a number and prints its multiplication table from 1 to 10. Use a **for** loop.

```
Enter a number: 7

7 x 1 = 7
7 x 2 = 14
7 x 3 = 21
...
7 x 10 = 70
```

8. Password Attempt System

Create a simple password verification program.
Requirements:
• Set a predefined password.
• Ask the user to enter the password.
• Allow a maximum of three attempts.
• If the password is correct, display **Login Successful**.
• If all three attempts fail, display **Account Locked**.
Use a **while** loop.

9. Student Attendance Tracker

A teacher wants to record attendance for five students.
Requirements:
• Create an array containing five student names.
• Use **foreach** to display each student's name.
• Display the total number of students.

```
Student Attendance List

1. Rahul
2. Priya
3. Akash
4. Sneha
5. Kiran

Total Students: 5
```

Part 4: Mini Project

10. Meter Serial Number Validator

An electricity department wants a console application to validate meter serial numbers before registering them.

Validation rules:

• Serial number must start with **MTR**.
• It must contain exactly 8 characters.
• The remaining 5 characters must be digits.
• Empty input should not be accepted.

Requirements:

1. Accept a meter serial number.
2. Validate its length.
3. Check whether it starts with MTR.
4. Check whether the remaining characters are digits.
5. Display whether the serial number is valid or invalid.
6. Allow the user to enter another serial number using a loop.

```
Enter Meter Serial Number: MTR12345

Serial Number: MTR12345
Status: Valid

Enter Meter Serial Number: ABC12345

Serial Number: ABC12345
Status: Invalid
```

Note: This question introduces string methods such as StartsWith() and char.IsDigit(). Students may need to learn these additional methods to complete it.